Rave Generation

PRESENTS

PMC60

USER MANUAL

1.0 Introduction

Welcome to Rave Generation: PMC60. PMC60 is a faithful digital emulation of the Lexicon PCM-60 Digital Reverberator, one of the most recognisable studio reverbs of the early 1980s. From compact rooms to bright plates, the original hardware became a staple because it delivered depth, density, and musical decay without getting in the way of a mix.
PMC60 recreates the unit at the microcode level. Instead of approximating the effect with a generic reverb algorithm or impulse responses, the plugin runs the original PCM-60 control-memory logic and ROM-derived program structures inside a cycle-aware CMU emulator. Modern conveniences such as Pre-delay with tempo sync, RT Extend, Tone shaping, and reverb ducking are added around that core so the plugin is easy to use in contemporary sessions while preserving the character of the hardware.

1.1 Key features

Rave Generation: PMC60 combines original-program authenticity with a small set of carefully chosen workflow enhancements:
- **ROM-driven reverb engine: Executes PCM-60 EPROM-derived program data through a control-memory-unit emulator rather than a generic plate or room algorithm.**
- **Eight original program banks: Room and Plate each provide four Size variations (Small, Medium Small, Medium Large, Large). Reverb Time controls decay length within the selected program.**
- **Authentic native clocking: Internal processing follows the original 24 kHz operating rate, with conversion to and from your DAW session rate.**
- **Hardware-matched filtering: Input anti-alias, DAC reconstruction, and output low-pass stages mirror the darker bandwidth of the original unit.**
- **Original-style contour controls: Bass and Treble shape decay behaviour and output tone in ways derived from the PCM-60 front panel.**
- **RT Extend: Pushes decay beyond the factory Reverb Time settings for longer ambient tails.**
- **Pre-delay: Adds up to 250 ms before the reverb input for cleaner separation between source and ambience.**
- **Pre-delay Sync: Locks the pre-delay time to the host tempo with musical note divisions from 1/2 down to 1/64T.**
- **Tone: A tilt EQ that shapes the signal entering the reverb tank, from dark to bright, independent of the original Bass and Treble contour switches.**
- **Ducking: Automatically attenuates the reverb tail when the dry input is loud, keeping the mix clean without external sidechain processing.**
- **Insert-friendly controls: Mix, Input Level, and Output Level make PMC60 practical both on aux sends and directly on tracks.**

2.0 User interface

The PMC60 interface follows the layout of the original PCM-60 while adding a compact control column for the extended plugin-only functions. From left to right, the panel is organised into utility controls, program selection, size and decay selection, and the Bass/Treble contour section.

2.1 Main controls

INPUT: Sets the level feeding the reverb engine from –20 dB to +6 dB. Keeping it near 0 dB delivers the cleanest behaviour; pushing it harder can drive the 16-bit signal path into audible clipping, much like the original hardware.

MIX: Blends dry and wet signal from 0% to 100%. At 100%, PMC60 behaves like a classic send-style reverb. Lower settings make it suitable for insert use.

OUTPUT: Applies final gain after dry/wet mixing, from –20 dB to +6 dB. Use it to match levels after changing programs or Mix settings.

BYPASS: Disables processing and passes audio through unchanged. Use this to compare the dry source with the active reverb setting.

2.2 Program selector

PLATE: Selects the Plate family. These programs are denser, brighter, and more diffuse, with a more open top end.

ROOM: Selects the Room family. These programs emphasise a more structured reflection pattern and a darker overall character.

Program choice also affects the Treble contour filter: Room uses a darker output contour than Plate. Changing Program reloads the selected bank and clears the reverb memory.

2.3 Size selector

SMALL: Tightest reverb field and shortest network lengths. Useful for small-room impressions, percussion, and subtle thickening.

MEDIUM SMALL: A versatile middle ground that works well on vocals, guitars, and general mix ambience.

MEDIUM LARGE: More spacious and open, with a wider early-reflection feel and longer internal delays.

LARGE: The biggest program size, suited to expansive rooms, plates, pads, and dramatic effect tails.

Each size selects a different ROM bank with its own topology of delays, allpasses, and output taps. Changing Size reloads the selected structure and clears the reverb memory.

2.4 Reverb Time selector

SHORT: Minimum factory decay for tight, quickly fading ambience.

MEDIUM SHORT: A balanced everyday setting for natural spaces and utility reverb.

MEDIUM LONG: A more sustained decay for featured reverb sounds.

LONG: Maximum factory decay within the selected Program and Size.

Unlike Program and Size, changing Reverb Time does not reload the algorithm. It changes the feedback gain inside the current reverb tank, so transitions can remain smooth during playback.

2.5 Contour controls

BASS: Adds low-frequency emphasis inside the feedback path. With Bass engaged, low frequencies decay more slowly, giving the tail more weight and warmth.

TREBLE: Activates high-frequency damping in the feedback path and an additional output contour low-pass filter. This makes the tail darken more naturally as it recirculates.

Together, Bass and Treble reproduce much of the PCM-60's signature tonal shaping: warm, slightly dark, and easy to place in a mix.

2.6 Additional controls

PRE-DELAY: Delays the signal feeding the reverb engine by up to 250 ms. This helps preserve the attack of vocals, snares, keys, and other transient-rich material before the reverb blooms.

RT EXTEND: Continuously increases feedback beyond the selected Reverb Time preset. At 0%, PMC60 stays within the factory range; at higher settings it moves into longer, more atmospheric decays.

PRE-DELAY SYNC: Click the metronome icon to lock the pre-delay to your DAW tempo. The knob switches from milliseconds to musical note divisions ranging from 1/2 down to 1/64T, including dotted and triplet values. Longer divisions that exceed the buffer length are clamped automatically.

TONE: A tilt EQ applied to the input signal before it enters the reverb tank. At 50% (centre) the EQ is flat. Turning left darkens the input by cutting highs and boosting lows; turning right brightens it. This is independent of the Bass and Treble contour switches, which shape the feedback and output instead.

DUCK AMOUNT: Sets how much the reverb tail is reduced when the dry input is loud. At 0% ducking is off. Higher values push the reverb further into the background during loud passages, letting it bloom back during quieter moments.

DUCK TIME: Controls how quickly the reverb returns to full level after the input drops. Lower values give a fast recovery; higher values keep the tail suppressed for longer.

2.7 Operational notes

For classic hardware-style routing, place PMC60 on an aux send and set Mix to 100% wet.

Program and Size changes clear the reverb memory. Reverb Time, Bass, Treble, RT Extend, Pre-delay, Tone, Duck Amount, Duck Time, Input, Mix, and Output can all be adjusted without reloading the program structure.

3.0 Technical overview

PMC60 is not a generic digital reverb with PCM-60-style voicing. Its sound comes from a ROM-driven architecture that recreates the original unit's program structures, delay behaviour, and bandwidth limits.

3.1 CMU microcode engine

The core of PMC60 is a cycle-aware emulator of the PCM-60's control-memory architecture:
- **Original program structures: Room and Plate banks are derived from the hardware's EPROM data.**
- **Fixed-point style behaviour: The reverb engine operates with 16-bit signed integer signal storage and a 16,384-sample DRAM buffer, matching the constraints of the original design.**
- **Mono-in, stereo-out topology: Stereo input is summed before entering the tank, while stereo width is created by the original output tap structure.**

3.2 Signal path

Audio passes through the following stages inside PMC60:
- **Input gain and optional pre-delay.**
- **Mono summing, followed by a 10.5 kHz anti-alias low-pass filter at the input.**
- **Conversion to the native 24 kHz processing rate used by the PCM-60 architecture.**
- **ROM-driven CMU reverb processing with program-dependent delays, allpasses, feedback paths, and output taps.**
- **DAC reconstruction filtering, output low-pass filtering, contour filtering when Treble is engaged, then dry/wet mix and final output gain.**

3.3 Reverb Time and RT Extend

Reverb Time in PMC60 is controlled by program-dependent feedback-gain tables derived from the original unit's published behaviour. Because each Size uses a different tank structure, the feedback values are size-dependent rather than global.
- **Base decay range: approximately 0.2 seconds at the shortest setting up to around 4.5 seconds at the longest factory setting, depending on Program and Size.**
- **Extended decay range: RT Extend pushes the feedback gain beyond the factory values and can reach roughly 7.1 seconds in Room and 7.4 seconds in Plate at the longest settings.**
- **Practical use: keep RT Extend low for authentic PCM-60 behaviour, or raise it for modern ambient tails and sound-design work.**

3.4 Frequency contouring

Bass and Treble do more than simple EQ. They change how the reverb evolves over time:
- **Bass introduces a low-frequency shelf inside the feedback loop centred around 800 Hz, so the low end decays more slowly.**
- **Treble applies feedback damping that progressively reduces high-frequency energy on each recirculation.**
- **Treble also engages an output contour low-pass filter: approximately 4.5 kHz for Room programs and 7 kHz for Plate programs.**

3.5 Analog signal-path emulation

PMC60 also models the bandwidth-limiting stages around the digital core:
- **Input anti-alias filter: a low-pass stage before conversion into the native reverb engine.**
- **DAC reconstruction filter: smooths the output of the low-rate processing stage.**
- **Output low-pass filter: completes the darker, slightly bandwidth-limited response associated with the original hardware.**

4.0 Installation & troubleshooting

4.1 System requirements

Before installing Rave Generation: PMC60, ensure that your system meets the following requirements:
- **Operating system: macOS 10.13 or later, or Windows 10 or later.**
- **Plugin formats: VST3 and AU.**
- **Processor: Intel Core i5 or equivalent.**
- **RAM: 4 GB minimum (8 GB or more recommended).**
- **Disk space: 100 MB free.**

4.2 Installation process

- **Download the installer from the official website or the platform where you purchased the plugin.**
- **Run the installer and follow the on-screen instructions.**
- **Launch your DAW and locate PMC60 in your plugin list.**
- **If prompted, activate the plugin using the licence key provided with your purchase.**

4.3 Troubleshooting

- **PLUGIN NOT SHOWING IN YOUR DAW: Check that the correct plugin path is enabled in your DAW's plugin manager, then rescan the plugin folders.**
- **ACTIVATION ISSUES: Verify your internet connection and enter the licence key exactly as provided.**
- **REVERB CUTS OUT WHEN CHANGING PROGRAM OR SIZE: This is normal. Those controls load a different ROM bank and clear the reverb memory.**

For technical support, updates, and additional information: ravegeneration.io